



August 25th, 2026

National Science Foundation
Program Directors, Pathways to Enable Open-Source Ecosystems (PESOSE), NSF 26-506

Subject: Letter of Collaboration — "RedTwin: AI-Enabled Digital Twins for Continuous Red Teaming of Open-Source Ecosystems" (Track 3)

Dear Program Directors,

My name is Buu Lam, and I am Director of Community Evangelism, NGINX and Open-Ecosystems, at F5, Inc., where a large part of my work centers on the NGINX open-source community. NGINX serves as the reverse proxy, load balancer, and web server in front of a very large share of the Internet's web applications, and my role is to connect the practitioners who run it in production with the engineers who build it. I write in support of the RedTwin proposal from Prof. Giovanni Vigna and Prof. Christopher Kruegel (UC Santa Barbara) and Prof. Adam Doupé and Prof. Tiffany Bao (Arizona State University).

I have no personal, professional, financial, or institutional relationship with any member of the proposing team. I am not a collaborator, co-author, former student, or advisor of any of the PIs, and neither I nor F5 has a financial interest in the proposed work. The team described the RedTwin approach to me because of my role in the NGINX community, and this letter reflects my own assessment in that capacity.

The questions NGINX users bring to me at conferences and in community forums are rarely of the form "is there a bug in this function." They are of the form "is my deployment safe." An NGINX instance almost never runs alone: it terminates TLS in front of a fleet of application servers, forwards to an identity provider, consults a cache, and sits inside a topology described by manifests and infrastructure-as-code that few individuals fully understand. The security posture a user actually experiences is a property of that whole composition, and today there is no practical, non-disruptive way for an operator to interrogate it. The community's answer has been folklore — hardening checklists, blog posts, and configuration snippets copied between deployments, with very little empirical grounding.

What I appreciate about RedTwin is that it offers a route out of that situation, and that the route depends on the software being open source. Replicating a deployment as a digital twin,



confining all aggressive analysis to the twin, and keeping every interaction with the production system passive is the first approach I have seen that could put deployment-security guidance on an evidentiary footing. I also find the proposed digital-twin competition a genuinely good idea: asking participants to build a representative twin from a description of an infrastructure teaches something that is hard to teach any other way, and it is an unusually accessible on-ramp for students and practitioners who want to learn infrastructure security by doing it.

My contribution to NGINX is community-facing. I run and support the channels through which NGINX practitioners exchange operational knowledge, I develop and deliver technical content and educational material about deploying NGINX correctly, and I carry community feedback back to the NGINX engineering organization.

I see several natural points of contact between this project and the community work I do, and I would welcome them, subject to the usual internal review at F5. The most valuable is probably access to deployment reality: research of this kind needs to know how NGINX is *actually* composed with other open-source components, and that knowledge sits with operators rather than in documentation. My role puts me in a position to help the team reach practitioners who might be willing to describe representative topologies. I would also expect to be a useful sounding board on findings — in particular on whether a given configuration-level result describes a pattern that is common in the field or an artifact of a synthetic test network, a distinction that determines whether a result deserves the community's attention. And should the project produce results and artifacts of practical value, there are established NGINX community channels, events, and technical-content programs through which they could reach operators rather than stopping at publication.

A collaboration of this kind would strengthen both sides. The researchers would gain access to the deployment reality of one of the most consequential pieces of open-source infrastructure in existence; the community would gain a rigorous way to reason about the security of the systems it builds, along with open-source tooling and students who understand infrastructure security. I am glad to support this proposal.

Sincerely,

Buu Lam
Director, Community Evangelism, NGINX, Open-Ecosystems
F5, Inc.

A handwritten signature in black ink, appearing to be 'Buu Lam', written over a horizontal line.

Signature